The documentation process conversion measures necessary to comply with this revision shall be completed by 1 October 1998

INCH-POUND

MIL-PRF-19500/157L
1 July 1998
SUPERSEDING
MIL-S-19500/157K
25 November 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE TYPES
1N941B, 1N943B, 1N944B, 1N945B, 1N946B, 1N941B-1, 1N943B-1, 1N944B-1, 1N945B-1,
1N946B-1, 1N941BUR-1, 1N943BUR-1, 1N944BUR-1, 1N945BUR-1, AND 1N946BUR-1
JAN, JANTX, JANTXV, AND JANS; RADIATION HARDENED (TOTAL DOSE ONLY)
JANTXVM, D, L, R, F, G, H, AND JANSM, D, L, R, F, G, AND H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 11.70 volts ± 5 percent, silicon, voltage-reference diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (DO-7 and DO-35) and figure 2 (DO-213AA).

1.3 Maximum ratings. Unless otherwise specified, maximum ratings (T_A = +25°C).

P_T 1/	T_{STG} and T_{op}	I_{ZM}	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-55 to +175	39	3.33

1/ For zero-TC voltage reference operation, I_Z is limited to 7.5 mA dc over -55°C to +150°C.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type	ΔV_Z (voltage - temperature stability) I_Z = 7.5 mA dc	Z_Z I_Z = 7.5 mA dc	V_Z I_Z = 7.5 mA dc		I_R V_R = 8.0 V
			Min	Max	
	mV dc	ohms	volts	volts	µA
1N941B, 1N941B-1, 1N941BUR-1	239.0	30	11.12	12.28	15.0
1N943B, 1N943B-1, 1N943BUR-1	47.0	30	11.12	12.28	15.0
1N944B, 1N944B-1, 1N944BUR-1	24.0	30	11.12	12.28	15.0
1N945B, 1N945B-1, 1N945BUR-1	12.0	30	11.12	12.28	15.0
1N946B, 1N946B-1, 1N946BUR-1	4.7	30	11.12	12.28	15.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

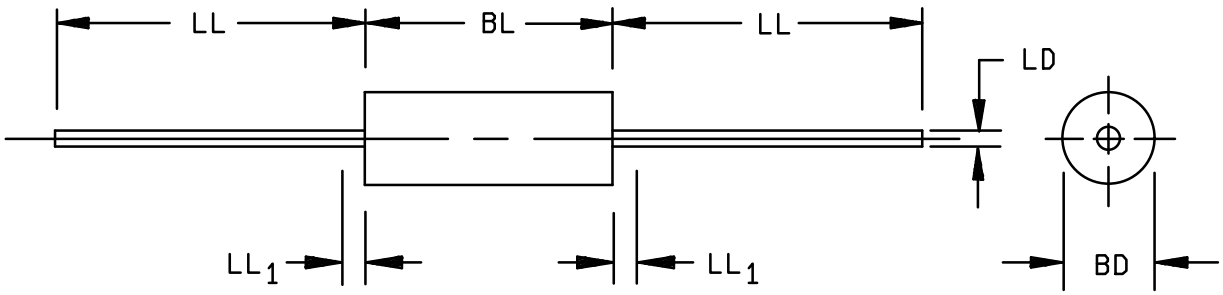
3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Dash-one construction. These devices shall be category I or II metallurgical bonding in accordance with MIL-PRF-19500.

3.4.3 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-PRF-19500, appendix A.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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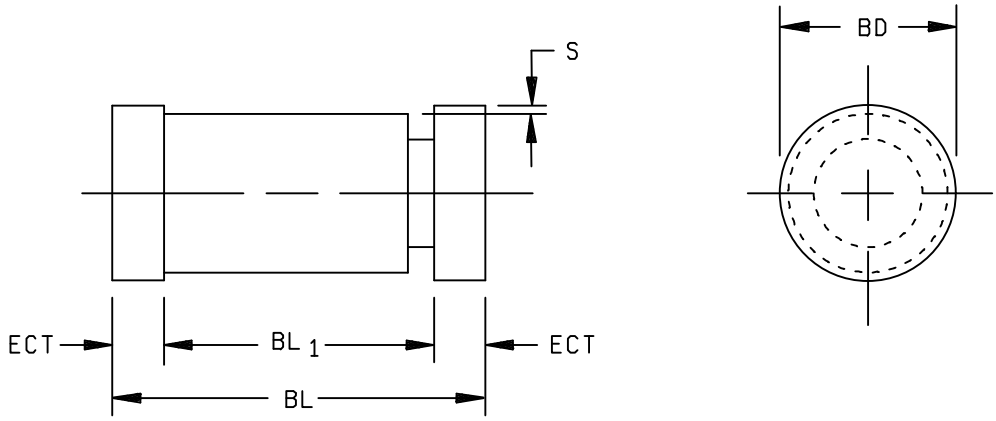


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	3
BL	.120	.300	3.05	7.62	3
LD	.018	.023	0.46	0.58	
LL	1.000	1.500	25.40	38.10	
LL_1	---	0.50	---	1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.

FIGURE 1. Physical dimensions, 1N941B, 1N943B, 1N944B, 1N945B, 1N946B, 1N941B-1, 1N943B-1, 1N944B-1, 1N945B-1, 1N946B-1 (DO-35 and DO-7).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
BL	.130	.146	3.30	3.70
BL_1	.100 REF		2.54 REF	
ECT	.016	.022	0.41	0.56
S	.001 Min		0.03 Min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, 1N941BUR-1, 1N943BUR-1, 1N944BUR-1, 1N945BUR-1, and 1N946BUR-1 (DO-213AA).

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3.5.1 Marking of "UR" version devices. For "UR" version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500, and table III herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table VI of MIL-PRF-19500	Measurement	
	JANS level	JANTX and JANTXV levels
11	V_Z, Z_Z, and I_R	V_Z, Z_Z, and I_R
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein, $\Delta Z_Z \leq \pm 15$ percent of initial reading. At T_A = +25°C, ± 2°C, $\Delta V_Z \leq \pm 8$ mV dc from initial value for 1N941B, 1N941B-1, 1N941BUR-1, 1N943B, 1N943B-1, 1N943BUR-1. $\Delta V_Z \leq \pm 6$ mV dc from initial value for 1N944B, 1N944B-1, 1N944BUR-1, 1N945B, 1N945B-1, and 1N945BUR-1. $\Delta V_Z \leq \pm 3$ mV dc from initial value for 1N946B, 1N946B-1, and 1N946BUR-1.	Subgroup 2 of table I herein, $\Delta Z_Z \leq \pm 15$ percent of initial reading. At T_A = +25°C, ± 2°C, $\Delta V_Z \leq \pm 8$ mV dc from initial value for 1N941B, 1N941B-1, 1N941BUR-1, 1N943B, 1N943B-1, 1N943BUR-1. $\Delta V_Z \leq \pm 6$ mV dc from initial value for 1N944B, 1N944B-1, 1N944BUR-1, 1N945B, 1N945B-1, and 1N945BUR-1. $\Delta V_Z \leq \pm 3$ mV dc from initial value for 1N946B, 1N946B-1, and 1N946BUR-1.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

I_Z = 7.5 mA dc, +0.75 mA dc, -0 mA dc, T_A = 150°C +5°C, -0°C.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1071	Test condition E.
B4	1037	I_Z = 7.5 mA dc at T_A = +150°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No forced air cooling on the device shall be permitted.
B5	1027	I_Z = 39 mA dc for 96 hours. T_A = +75°C or adjusted as required to give an average lot . T_J = +200°C.
B6		Not applicable

4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1071	Test condition E.
B3	1027	T_A = +150°C and I_Z = 7.5 mA dc. NOTE: When group C (subgroup 6) is required, group B (subgroup 3) is not required for that lot.
B6		Not applicable

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500, and herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Lead tension: 4 pounds weight, t = 15 ± 3 seconds; test condition A. Not applicable to "UR" version devices. Lead fatigue: Test condition E, not applicable to "UR" version devices.
C2	1071	Test condition E.
C6	1026	T_A = +100°C and I_Z = 7.5 mA dc (see 4.5.2).

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500 and table II herein. MIL-PRF-19500, appendix F is not required when homogeneous product of radiation hard assembly (in accordance with applicable 36D form) is submitted and successfully passes group D sampling of total dose at an outside facility compliant to MIL-STD-750, method 1019. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die, as also described in the submitted 36D forms.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

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4.5.1 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z per diode type at I_Z = 7.50 ± 0.01 mA dc.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing, at 340 hours, and at the conclusion of the life test. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Reference-voltage time stability		T_A = +100°C ± 2°C (see 4.5.2 and 4.5.3) I_Z = 7.5 ± 0.01 mA dc	ΔV_Z			
1N941B, 1N941B-1, 1N941BUR-1		(0 to 340 hours)			7	mV dc
1N943B, 1N943B-1, 1N943BUR-1					7	mV dc
1N944B, 1N944B-1, 1N944BUR-1					6	mV dc
1N945B, 1N945B-1, 1N945BUR-1					5	mV dc
1N946B, 1N946B-1, 1N946BUR-1					4	mV dc
1N941B, 1N941B-1, 1N941BUR-1		(340 to 1,000 hours)			4	mV dc
1N943B, 1N943B-1, 1N943BUR-1					4	mV dc
1N944B, 1N944B-1, 1N944BUR-1					3	mV dc
1N945B, 1N945B-1, 1N945BUR-1					3	mV dc
1N946B, 1N946B-1, 1N946BUR-1					2	mV dc

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 seconds minimum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between .375 inch (9.53 mm) and .500 inch (12.70 mm) from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reference voltage (see 4.5.3)	4022	I_Z = 7.5 ±0.01 mA dc	V_Z	11.12	12.28	V dc
Small-signal breakdown impedance	4051	I_Z = 7.5 ±0.1 mA dc, I_{sig} = 0.75 mA ac	Z_Z		30	Ω
Reverse current leakage	4016	DC method; V_R = 8.0 V dc	I_R		15	µA
Subgroup 3						
Voltage-temperature stability (see 4.5.1 and 4.5.3)		I_Z = 7.5 ± 0.01 mA dc, T_A = -55°C, ± 0°C. 25°C, +75°C, +150°C ± 2°C	ΔV_Z			
1N941B, 1N941B-1, 1N941BUR-1					239.0	mV dc
1N943B, 1N943B-1, 1N943BUR-1					47.0	mV dc
1N944B, 1N944B-1, 1N944BUR-1					24.0	mV dc
1N945B, 1N945B-1, 1N945BUR-1					12.0	mV dc
1N946B, 1N946B-1, 1N946BUR-1					4.7	mV dc
Subgroups 4, 5, and 6						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		ΔV_{Z1} pre-post irradiation change		Unit
	Method	Conditions		M, D, R, L, F, G, and H		M, D, R, L, F, G, and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = 7.5 ± 0.01 mA dc, condition A								
Reference voltage (see 4.5.3)	4022	I_Z = 7.5 ± 0.01 mA dc	V_Z	11.12	12.28	11.12	12.28			V dc
Small-signal breakdown impedance	4051	I_Z = 7.5 ± 0.01 mA dc I_{sig} = 0.75 ac	Z_Z		30		30			ohms
Reverse current leakage	4016	DC method; V_R = 8.0 V dc	I_R		15.0		15.0			µA
Voltage stability (4.5.3)	4022	I_Z = 7.5 ± 0.01 mA dc, T_A = +25°C, ± 2°C	ΔV_Z							
1N941B, 1N941B-1 1N941BUR-1									± 5.0	mV
1N943B, 1N943B-1 1N943BUR-1									± 4.0	mV
1N944B, 1N944B-1 1N944BUR-1									± 3.0	mV
1N945B, 1N945B-1 1N945BUR-1									± 2.0	mV
1N946B, 1N946B-1 1N946BUR-1									± 1.5	mV

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation.

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TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan 1/
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock temperature cycling	1051	500 cycles	
Electrical measurement		See table I, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state dc life	1038	Condition B, 1,000 hours	
Electrical measurement		See table I, subgroup 2	
Subgroup 3			
Not applicable			
Subgroups 4, 5, and 6			
Not applicable			

1/ For sampling plans not specified, see MIL-PRF-19500.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.5 Substitution of devices. Device types within this series with higher type numbers (lower ΔV_Z voltage-temperature stability) are a direct oneway substitution for lower type numbers (higher ΔV_Z voltage-temperature stability).

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, SM
Navy - AS, CG, MC
Air Force - 19, 80, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1970)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/157L	**2. DOCUMENT DATE (YYMMDD)** 980701

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE TYPES 1N941B, 1N943B, 1N944B, 1N945B, 1N946B, 1N941B-1, 1N943B-1, 1N944B-1, 1N945B-1, 1N946B-1, 1N941BUR-1, 1N943BUR-1, 1N944BUR-1, 1N945BUR-1, AND 1N946BUR-1, JAN, JANTX, JANTXV, AND JANS; RADIATION HARDENED (TOTAL DOSE ONLY) JANTXVM, D, L, R, F, G, H, AND JANSM, D, L, R, F, G, AND H

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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